TRUNKBELAY · REGULATORY AND STANDARDS REGISTER

TrunkBelay regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First register (kit 1.9.1 upgrade), complete for stages P0 to P3

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TrunkBelay is published as an open engineering reference, not as certified rescue or fall-protection equipment, and it is not placed on any market. Its bought rescue gear is specified to recognised standards (G3 to G6), and those certificates are what a user relies on. The made collar is an anchor for a rescue, and no certification route for a home-built palm anchor is known: on paper it would not pass the 12 kN static test of EN 795 (G2), which is the main gap if a partner ever wants a certified product. In India, where the first co-design partner is sought, no mandatory standard for rope rescue equipment was found; that is to assess with the partner and a qualified reviewer (G8, G9). Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Regulation (EU) 2016/425 on personal protective equipment	European Union	Placing PPE on the EU market: risk categories, conformity assessment and CE marking	Only if a kit were ever sold or supplied in the EU; an open design shared without being placed on the market is outside it	Project lead	To assess (does not apply while nothing is placed on the market)
G2	EN 795:2012, protection against falls from a height, anchor devices	European Union; reference elsewhere	Anchor devices for one or more users, types A to E; static strength test at 12 kN (metallic) or 18 kN (textile without a service-life marking), plus dynamic tests	The collar acts as a transportable anchor (type B class). Sized for 2.5 kN static (PR-001) with an estimated breaking load of about 5.7 kN, it would not pass the 12 kN static test	Design lead	Certification needed later; on paper not met (TKB-STM-001 S30; TKB-DEC-001 O7)
G3	EN 341:2011, descender devices for rescue	European Union; reference elsewhere	Rescue descenders: classes A to D by descent energy (class A 7.5 MJ), descent speed 0.5 to 2 m/s, holding force at most 200 N, touch surfaces at most 48 C	The bought auto-locking descender (BOM line 17) is specified as EN 341 class A or EN 12841 type C or equal	Design lead	Applies when bought (ask for the declaration of conformity; PR-011, PR-015)
G4	EN 362:2004, connectors	European Union; reference elsewhere	Carabiners and connectors for fall protection; class B (basic) connectors	The three locking carabiners (BOM line 16)	Design lead	Applies when bought (PR-015)
G5	EN 1891:1998, low stretch kernmantle ropes	European Union; reference elsewhere	Type A ropes 8.5 to 16 mm: at least 22 kN, 15 kN for 3 minutes with figure-eight terminations, elongation and sheath slippage limits	The 11 mm lowering rope (BOM line 18)	Design lead	Applies when bought (PR-015; TKB-STM-001 S12)
G6	EN 1498:2006, rescue equipment, rescue loops	European Union; reference elsewhere	Rescue loops and evacuation triangles, classes A to C	The evacuation triangle, class B (BOM line 19)	Design lead	Applies when bought (PR-012, PR-015)
G7	EN 818-2, short link chain for lifting purposes, grade 8	European Union; reference elsewhere	Dimensions, working load limits and test forces of grade 8 lifting chain	The 6 mm grade 80 chain and its connecting and master links (BOM lines 11 to 14): 7.8	Design lead	Applies when bought (ask for the certificate;

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
				mm minimum link opening, 45.2 kN minimum breaking force (TKB-STM-001 S11, S27)		TKB-DEC-001 item 2)
G8	Indian Standards for fall protection equipment, for example IS 3521 (industrial safety belts and harnesses)	India	Bureau of Indian Standards specifications for safety harnesses and belts	Whether any Indian Standard or Quality Control Order covers a rescue anchor, descender or kit used by coconut climbers is not known	Co-design partner	To assess
G9	Occupational safety law for agricultural and plantation work, and the Coconut Development Board's climber training and insurance schemes	India (Kerala and Tamil Nadu first)	Duties of employers and schemes for climbers; training content of the Friends of Coconut Tree scheme	Whether the drill could be added to the training scheme, and whether the scheme's insurance has conditions on rescue equipment	Co-design partner	To assess
G10	NFPA 2500, operations and training for technical search and rescue incidents and life safety rope and equipment for emergency services	United States; reference elsewhere	Life safety rope, harnesses and auxiliary equipment for emergency services (it absorbed the former NFPA 1983)	The bought descender class is also listed to NFPA 2500 (Petzl catalogue); a reference for fire services elsewhere	Design lead	Does not apply (reference only)
G11	ANSI/ASSP Z359 family, fall protection, including Z359.4 assisted-rescue and self-rescue systems	United States; reference elsewhere	Rescue systems and components for work at height	Reference for rescue practice; the kit is not marketed in the United States	Design lead	Does not apply (reference only)
G12	Patents: 3M DBI-SALA Cynch-Lok ratcheting cinch geometry; CA2681870C (Bashlin, lapsed 2015); WO2012131439A1 (ceased)	International	Freedom to operate	Design-around kept (PR-017): no ratchet, cam or cleated gripping plate (TKB-PRB-001, TKB-PRC-001)	Project lead	Met by design (preliminary screen, not legal advice)

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- SATRA. "EN 341:2011 Descender Devices for Rescue." https://www.satra.com/spotlight/article.php?id=439 (checked 2026-10-25) (G3).

- Vandernet. "EN 795 Standard Explained." https://www.vandernet.com/tuotetiedostot/EK-FPPFM30x/EN_795_standard_explained.pdf (checked 2026-10-25) (G2).
- Vandernet. "EN 1891 Standard Explained." https://vandernet.com/tuotetiedostot/EK-FPPFM20x/EN_1891_standard_explained.pdf (checked 2026-10-25) (G5).
- S3i. "Lifting Chain, Grade 80, Alloy Steel." https://www.s3i.co.uk/lifting-chain-grade-80.php (checked 2026-10-25) (G7).
- Petzl. "Professional catalogue 2026: I'D S certifications." https://catalogs.petzl.com/Petzl-pro-catalog-2026/en/68-Rope-Clamps.html (checked 2026-10-25) (G3, G10).
- Piatt, Lance. "Rope Rescue Safety Factors." Rigging Lab Academy, July 3, 2023. https://rigginglabacademy.com/rope-rescue-safety-factors/ (checked 2026-10-25) (G2, safety factor practice).
- Regulation (EU) 2016/425, EN 362:2004, EN 1498:2006, IS 3521, NFPA 2500 and ANSI/ASSP Z359.4 are cited by number and title only; their texts were not consulted for this register, and each applicability is to be confirmed by the role named.
- Patent references as screened in TKB-PRB-001 (G12).